

APPENDIX G

HIGH PRESSURE NATURAL GAS INSTALLATIONS

G.1 General. This appendix addresses natural gas distribution piping requirements for systems where the gas pressure is at or above 15 psig (103.4 kPa gauge). Installations of gas piping at pressures at or above 15 psig (103.4 kPa gauge) and equipment and appliances using gas with an inlet pressure at or above 15 psig (103.4 kPa gauge) shall be considered as high pressure natural gas installations.

G.1.1 Fire Department approval. High pressure natural gas installations shall be approved by the Fire Department. All design documents associated with the installation shall be submitted to the Fire Department for approval. The Fire Department shall witness and approve final testing of the installation.

G.1.2 Certificate of fitness. High pressure natural gas installations shall be operated under the supervision of a person holding a Certificate of Fitness issued by the Fire Department when required by the *New York City Fire Code*.

G.2 Construction requirements. Buildings with high pressure natural gas installations shall meet the requirements of this section.

G.2.1 Structural requirements. The structural integrity of the building shall meet the requirements of Section 1615.6 of the *New York City Building Code* for gas explosions. An explosion analysis shall be conducted where the gas pressure exceeds 125 psig (861.8 kPa gauge). Where an explosion analysis shows explosion pressure exceeding 430 psf (20.5 kPa gauge), the building's structural integrity shall be maintained at pressure levels determined by the explosion analysis. The explosion scenario and explosion analysis shall be approved by the Fire Department.

G.2.2 Fire-resistance-rated[‡] rooms and spaces. Rooms and spaces containing high pressure natural gas piping shall be separated from all other areas of the building by fire barriers or horizontal assemblies, or both, having a fire-resistance rating of not less than 3 hours.

G.2.2.1 Shaft requirements. Vertical runs of high pressure gas piping within a building shall be enclosed in masonry shafts constructed of walls not less than 4 inches (101.6 mm) in thickness and sealed to prevent any gas leakage from the shaft. Such shaft shall be vented to the outdoors at the top. Such shaft shall not be located adjacent to an exit stairway or exit passageway unless the shaft wall separating the exit stairway or exit passageway from the shaft is designed to resist a potential gas explosion in accordance with Section 1615.6 of the *New York City Building Code*.

G.2.3 Automatic sprinkler system. Buildings and structures shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 of the *New York City Building Code*, except where an alternative fire-extinguishing system is approved in accordance with the *New York City Fire Code* and the *New York City Building Code*.

G.2.4 Gas detection. Rooms and spaces containing high pressure gas piping must be equipped with an approved and listed gas detection alarm system.

G.2.4.1 Gas detection system. A gas detection system including placement of gas detectors shall be installed in accordance with the manufacturer's recommendations, its listing and Section 908 of the *New York City Building Code*.

G.2.4.2 Supervision. Gas detectors shall be supervised by a building fire alarm system in buildings where fire alarm systems are required or installed.

G.2.4.3 Power supply. Power supply to the system, wiring of the system, its associated components and outputs shall be in accordance with NFPA 72 as modified by Appendix Q of the *New York City Building Code* and Section 907 of the *New York City Building Code*.

G.2.4.4 Alarm. An audible and visual alarm shall be provided at the gas detection panel with an audible silence switch only.

G.2.4.5 Alarm transmission. The gas detection system shall transmit a trouble signal and an alarm to the supervising station and to a 24-hour supervised location within the building.

G.2.4.6 Shutoff valves closure. The gas detection system alarm activation shall trigger emergency gas shutoff valves serving the affected area to close and stop the gas flow.

G.2.5 Ventilation. Rooms containing appliances or equipment operating with gas pressure at or above 15 psig (103.4 kPa gauge) shall be provided with ventilation in accordance with the requirements of Chapter 4 of the *New York City Mechanical Code*.

G.2.5.1 Explosion prevention. For rooms or spaces containing appliances operating with gas pressure at or above 15 psig (103.4 kPa gauge) and less than 125 psig (861.8 kPa gauge), explosion prevention systems shall be provided in accordance with NFPA 69.

G.3 High hazard. Rooms and areas containing appliances and equipment operating with gas pressure at or above 125 psig (861.8 kPa gauge) shall be classified as a Group H-2 occupancy and shall comply with the requirements of this section.

G.3.1 Explosion venting. Appliances and equipment using gas at or above 125 psig (861.8 kPa gauge) shall be located in rooms provided with explosion venting in accordance with NFPA 68.

G.3.1.1 Gas turbines. Gas turbine installations shall comply with the requirements of NFPA 37 and the requirements of Title 28 of the *Administrative Code*.

G.3.1.2 Gas turbine rooms. For rooms containing gas turbines and fuel gas compressors, the design of the explosion venting system shall be based on the explosion resulting from the lower explosion limit being achieved within the room housing the turbine or fuel gas compressor, including its enclosure. All control valve stations, filters, and related accessories shall be placed in the gas turbine room or a separately protected room.

G.3.2 Emergency ventilation. An emergency ventilation system shall be provided in accordance with NFPA 69 and shall be approved by the Fire Department. Ventilation calculations as listed in Annex D Ventilation Calculations of NFPA 69 shall be provided, and the gas release rate shall be approved.

G.3.3 Electrical equipment. Electrical equipment within rooms and enclosures requiring emergency ventilation shall conform to the *New York City Electrical Code* for Class 1 Division 2 requirements.

G.4 Gas meter room. A gas meter room served by gas at a pressure at or above 15 psig (103.4 kPa gauge) shall meet the requirements of the *New York City Electrical Code* for Class 1 Division 2 requirements, and shall comply with Appendix E of this code.

G.5 Piping requirements. High pressure gas piping shall comply with the requirements of this section.

†††**G.5.1 Piping material.** Installations of natural gas piping operating at pressures of 125 psig (861.8 kPa gauge) and above shall comply with the requirements of ASME B31.1.

G.5.2 Double wall piping. Horizontal piping that traverses within a building from a protected room or shaft to a protected room or shaft shall be run in an outer pipe of the same pressure rating as the inner pipe. The outer pipe shall be welded and the annular space between the inner and outer pipe shall be equal to or greater than the inside diameter of the inner pipe. The annular space shall be monitored for natural gas with an approved gas detection alarm system and shall be vented to the exterior of the building. The outer pipe must open to the shaft and/or protected room or to the outside air.

G.5.3 Piping identification. Piping shall be identified with markings in accordance with ASME A13.1.

G.5.4 Emergency gas shutoff. An emergency gas shutoff valve shall be provided on the gas supply outside of any room containing an appliance utilizing gas at a pressure at or above 15 psig (103.4 kPa gauge). The emergency shutoff valve shall be controlled from a break glass station located outside the room served and from the gas detection system monitoring the room. The emergency shutoff valve shall automatically stop the gas flow to the room(s) containing the appliance(s) in the event of an unsafe condition. The emergency shutoff valve shall be manually operable. Emergency gas shutoff valve bypasses shall be prohibited.

G.6 Special inspection required. The entire high pressure natural gas installation, including piping, equipment, appliances, gas detection and control systems, shall be subject to special inspection as set forth in Section 1705.24 of the *New York City Building Code*.

G.7 Cleaning and purging procedures. Cleaning and purging procedures for high pressure natural gas piping, equipment and appliances shall be in accordance with NFPA 56.